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Task: Implementation of FCFS algorithm.

Problem Name:

Find the waiting time, average time and turnaround time for a given process of a system FCFS scheduling algorithm.

Solution:

```
#include <bits/stdc++.h>

using namespace std;

#define int long long
#define M 1000000007
#define N 1000000
#define endl "\n"
#define all(v) v.begin(), v.end()
#define forn for (int i = 0; i < n; i++)
#define yes cout << "YES" << endl
#define no cout << "NO" << endl

void marwan()
{
    int n;
    cout << "Enter number of processes: ";
    cin >> n;
    vector<int> burst(n), waiting(n), turnaround(n);
    cout << "Enter burst time for each process:\n";
    for (int i = 0; i < n; i++)
    {
        cout << "Process " << i + 1 << ": ";
        cin >> burst[i];
```

```

}

waiting[0] = 0;
for (int i = 1; i < n; i++)
{
    waiting[i] = waiting[i - 1] + burst[i - 1];
}
for (int i = 0; i < n; i++)
{
    turnaround[i] = waiting[i] + burst[i];
}

double totalWT = 0, totalTAT = 0;
cout << "\nProcess\tBurst\tWaiting\tTurnaround\n";
for (int i = 0; i < n; i++)
{
    cout << "P" << i + 1 << "\t"
        << burst[i] << "\t"
        << waiting[i] << "\t"
        << turnaround[i] << "\n";
    totalWT += waiting[i];
    totalTAT += turnaround[i];
}

cout << fixed << setprecision(2);
cout << "\nAverage Waiting Time = " << totalWT / n;
cout << "\nAverage Turnaround Time = " << totalTAT / n << endl;
}

int32_t main()
{
    ios_base::sync_with_stdio(false);
    marwan();
    return 0;
}

```

}

Sample Input:

Enter the number of processes: 5

Enter the burst time of process P1: 10

Enter the burst time of process P2: 5

Enter the burst time of process P3: 3

Enter the burst time of process P4: 9

Enter the burst time of process P5: 12

Sample Output:

Process Burst Time Waiting Time Turnaround Time

P1 10 0 10

P2 5 10 15

P3 3 15 18

P4 9 18 27

P5 12 27 39

Average Waiting Time = 14.00

Average Turnaround Time = 21.80